Assignment 5 – Describe your data and one or more of your research questions from the point of view of measurement theory, guided by Figure 1 in Voje et al. 2023. Describe the theoretical context, relevant attributes, what you will measure on what kind of scale, and how this will affect your statistical analyses. Also, for at least one comparison, discuss how you will decide whether a difference is important – a cutoff level, or describe how you will investigate this question further.

As a quick reminder, my dataset has information on the water quality and stream characteristics (dissolved carbon/nitrogen, periphyton biomass) of twelve streams in the Gaspesie Peninsula which have been impacted by spruce budworm defoliation over the last decade. Because of their different geographic features and management interventions by the Canadian government, the severity of outbreak varies among streams, creating a gradient on which we can compare streams. My main question was whether the level of SBW defoliation impacted the nutrient or periphyton levels within the streams.

The theoretical context of my data is measuring the difference in stream quality across sites and years in the Gaspesie region. The selected attributes for this study are the concentration of dissolved nutrients (organic/inorganic carbon, nitrogen, and phosphorous; measured in mg/L) and biomass of periphyton (cyanobacteria, green algae, diatoms; measured in ug/cm^2). The data falls into the ratio category described by Voje et al. (2023) as the specific numerical values have meaning (and cannot be replaced by a character/symbol) and the value of 0 means there is nothing of the entity in question. Because of my type of data, I am able perform most types of statistics, including logarithmic transformation and taking means and variances, as well as using statistical methods that rely on these calculations.

For the second part of the assignment, I have determined that I will use proportional differences to determine whether periphyton biomass changes in a single site over multiple years are "important". While I may be able to see small, absolute changes statistically, this may not be ecologically relevant, especially when considering the overall function of the ecosystem. I believe a proportional change (perhaps a change greater than 50% from one year to the next) may better reflect biologically important shifts in periphyton success. This 50% threshold would be provisional, and a more appropriate threshold could be determined by a deeper literature review into periphyton and its role in the ecosystem.